

Introduction to Program Synthesis (SS 2026)

Chapter 2.4 - Foundations (Program Representation: Grammar)

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Computer Programs: Representations

Grammar-based representation

- ▶ **Grammar** → set of rules commonly used to describe the syntax of sentences and expressions (in a certain language)
- ▶ **Context Free Grammar (CFG)** → used to describe context-free languages (CFL)
 - ~ CFL → used for many programming languages
 - ~ Production rules can be applied to non-terminal symbols regardless of the context (i.e. surrounding symbols)
- ▶ CFG can be used to represent computer programs
 - ~ CFG based representation models are used for program synthesis
 - ~ **Grammar-guided program synthesis**

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Definition (Context Free Grammar)

A context free grammar (CFG) can be defined a four-tuple $\{V, \Sigma, P, S\}$:

- ▶ $V \rightarrow$ finite set of non-terminal symbols or variables
- ▶ $\Sigma \rightarrow$ finite set of terminals
- ▶ $P \rightarrow$ set of production rules
 $\leadsto$ Finite relation in $V \times (V \cup \Sigma)^*$
- ▶ $S \in V \rightarrow$ starting symbol

A grammar is context free if every production R is in form $R \rightarrow (V \cup \Sigma)^*$ with $G \in V$

- ▶ Contains only substitution rules in which exactly one non-terminal symbol is always derived from an arbitrarily long sequence of non-terminal and terminal symbols

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Example (Context-free Grammar)

```
1 S ⇒ <exp>
2 <exp> ⇒ <var> | <exp><op><exp>
3 <op> ⇒ + | - | * | /
4 <var> ⇒ x
```

Listing: Simple context-free grammar

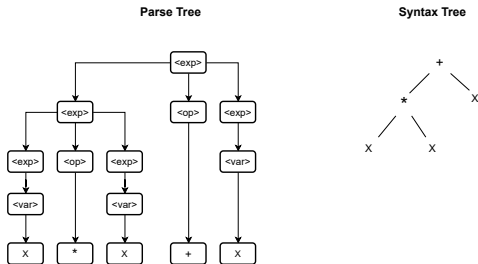


Figure: The expression $x * x + x$ that can be produced by the CFG

Computer Programs: Representations

Grammar-based representation

- ▶ Integer-based encoding of programs → enables a simple and straight-forward way to variate programs
- ▶ **Grammatical Evolution** [RCO98] → grammar-based program synthesis with artificial evolution mechanisms
- ▶ Encodes and decodes programs represented as grammar

Computer Programs: Representations

Grammar-based representation

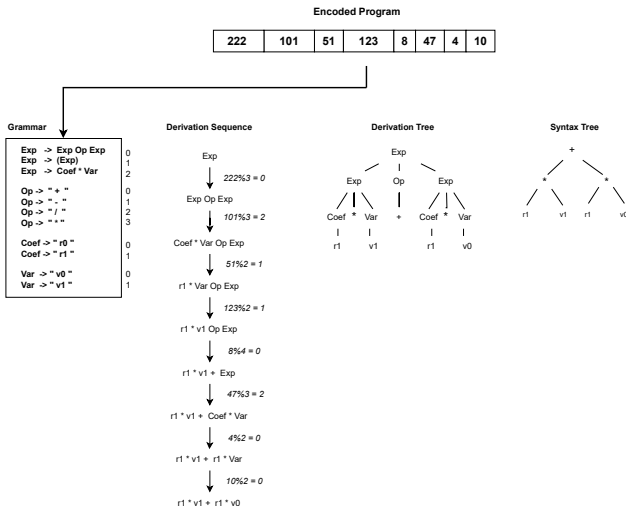


Figure: Grammar-based encoding of programs

References

- [RCO98] C. Ryan, J. J. Collins, and M. O'Neill. "Grammatical Evolution: Evolving Programs for an Arbitrary Language". In: *Genetic Programming, First European Workshop, EuroGP'98, Paris, France, April 14-15, 1998, Proceedings*. Vol. 1391. Lecture Notes in Computer Science. Springer, 1998, pp. 83–96.